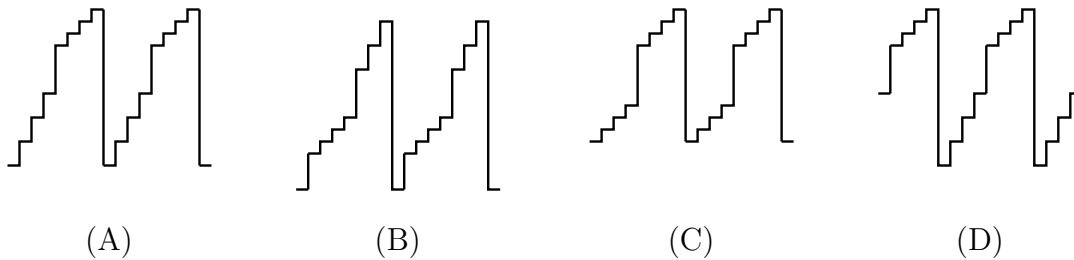
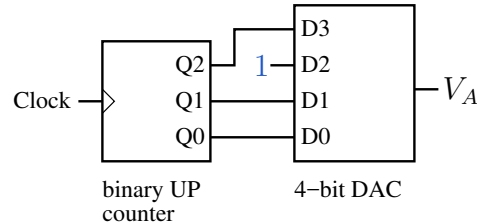


Basic Electronics

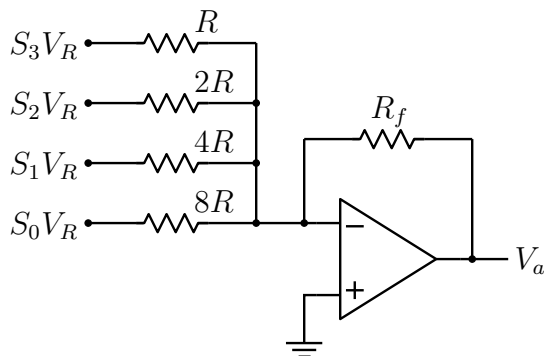
Instructor: M. B. Patil, IIT Bombay
Assignment for Week 12

1. For the circuit shown in the figure, which waveform correctly describes $V_A(t)$?

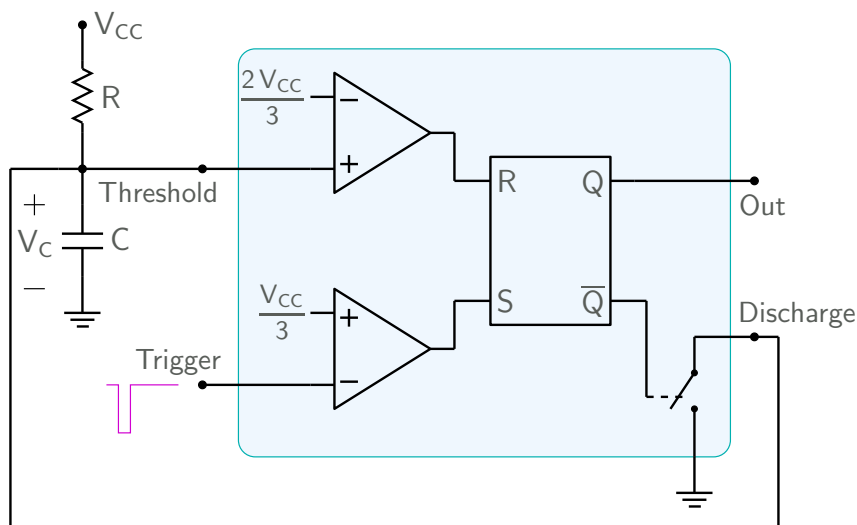


2. The full-scale input for an ADC is 10 V. The minimum resolution required is 5 mV. What is the minimum number of bits required for the ADC?
- (A) 9 (B) 10 (C) 11 (D) 12
3. In what way a flash ADC is superior to other ADC types?
- (A) It has a high resolution.
 (B) It offers a high conversion speed.
 (C) It is more immune to noise.
 (D) It has a low component count.
4. The resolution of an 8-bit DAC is 39 mV. What would be the output voltage for an input of 0110 1110?
- (A) 4.29 V (B) 3.56 V (C) 6.27 V (D) 3.85 V
5. A 10-bit ADC gives 10 1000 1000 when an input voltage of 3.164 V is applied. The full-scale input of the ADC is
- (A) 10 V (B) 15 V (C) 8 V (D) 5 V
6. When an input voltage of 1.785 V is applied to the ADC of Q-5, what output is expected?
- (A) 00 1010 1100 (B) 01 0110 1101 (C) 01 1001 1110 (D) 00 1001 1100
7. A 10-bit counter type ADC operates with a clock frequency of 1 MHz. What is the maximum number of samples it can process per second?
- (A) 765 (B) 1240 (C) 976 (D) 1124

8. In the weighted-resistor DAC shown in the figure, $R_f = R$, $V_R = -5\text{ V}$, and the op-amp can be considered to be ideal. Assume the resistance values to be perfect, i.e., exactly as shown in the figure. What is V_a for $S_3S_2S_1S_0 = 1011$?

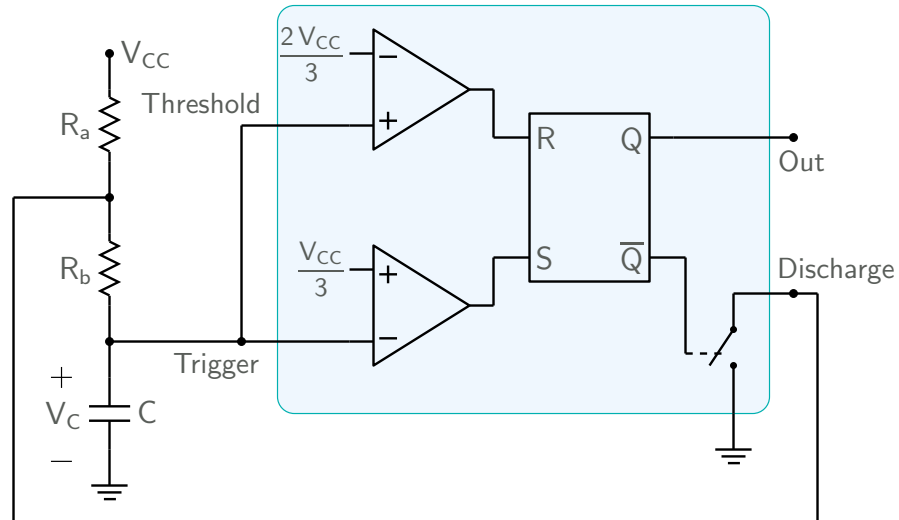


- (A) 6.25 V (B) 6.75 V (C) 6.875 V (D) 7.25 V
9. In the DAC of Q-8, suppose all resistances have a 2% tolerance. For $S_3S_2S_1S_0 = 1011$, what is the worst-case deviation (magnitude) in V_a from its expected value?
- (A) 0.16 V (B) 0.28 V (C) 0.22 V (D) 0.08 V
10. In the DAC of Q-8, we want to limit the deviation in V_a from its expected value for $S_3S_2S_1S_0 = 1011$ to 0.2%. What is the minimum tolerance required for the resistances?
- (A) 0.1% (B) 0.2% (C) 0.05% (D) 0.25%
11. A 10-bit counter type ADC operates with a clock frequency of 1 MHz. The worst-case conversion time per sample is about
- (A) 1 msec (B) 0.5 msec (C) 10 μsec (D) 0.25 μsec
12. In the monostable multivibrator shown in the figure, $V_{CC} = 5\text{ V}$, and $C = 50\text{ }\mu\text{F}$. When a negative-going trigger is applied, we want to get a 1-sec output pulse. What value of R is appropriate?



- (A) 22 k Ω (B) 26.9 k Ω (C) 15.5 k Ω (D) 18.2 k Ω

13. When a negative-going trigger is applied to the monostable multivibrator of Q-12, V_C varies between
 (A) 0 V and 5 V (B) 0 V and 3.33 V (C) 1.67 V and 3.33 V (D) 0 V and 1.67 V
14. In the astable multivibrator shown in the figure, let T_H denote the interval of high V_o and T_L denote the interval of low V_o . Which of the following statements is correct?



- (A) V_C varies between 0 V and $2V_{CC}/3$.
 (B) T_L increases as R_a is increased.
 (C) T_L increases as R_b is increased.
 (D) T_H is independent of R_b .